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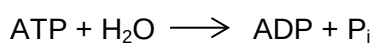
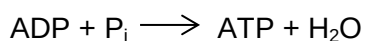
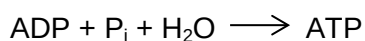
Scientists investigated treatment of a human bladder infection caused by a species of bacterium. This species of bacterium is often resistant to the antibiotics currently used for treatment.

They investigated the use of a new antibiotic to treat the bladder infection. The new antibiotic inhibits the bacterial ATP synthase enzyme.

0 5 . 1

Place a tick (✓) in the appropriate box next to the equation which represents the reaction catalysed by ATP synthase.

[1 mark]

☐☐☐☐

0 5 . 2

The new antibiotic is safe to use in humans because it does **not** inhibit the ATP synthase found in human cells.

Suggest why human ATP synthase is not inhibited and bacterial synthase is inhibited.

[1 mark]

Question 5 continues on the next page

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05 . 3

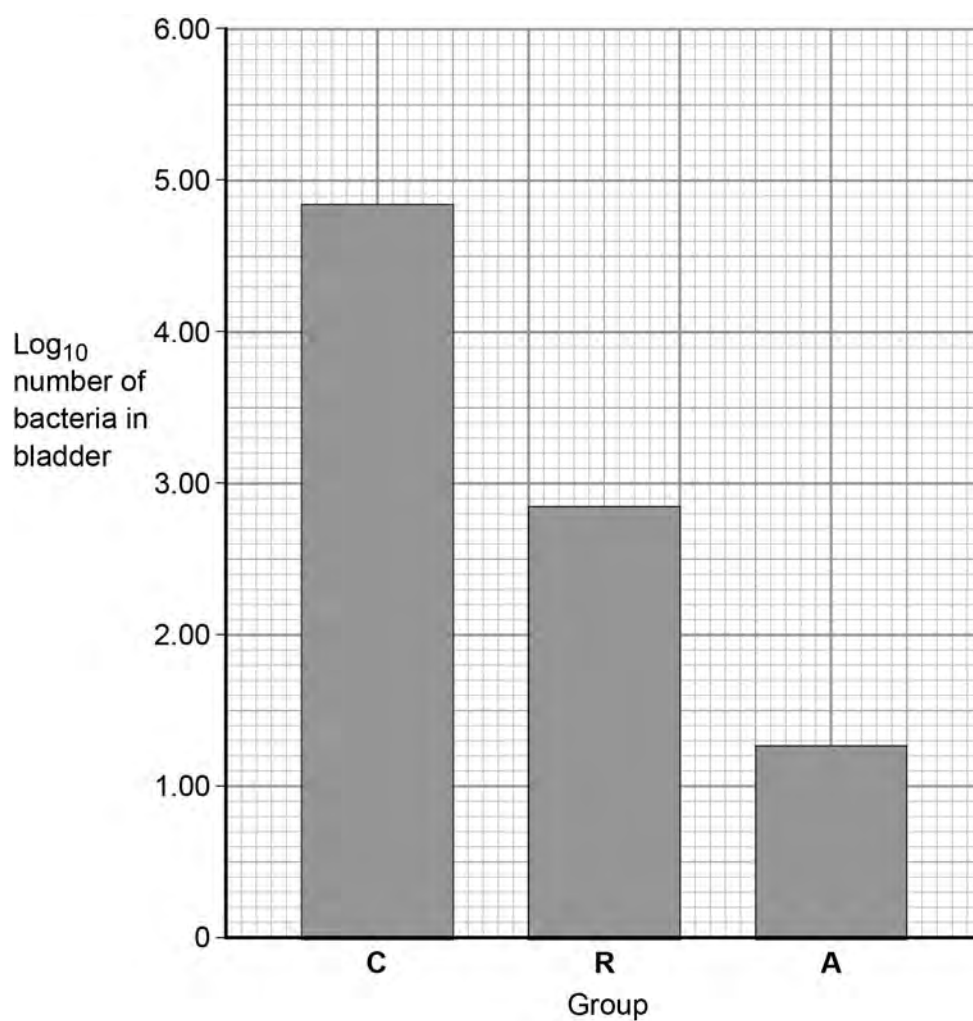
The scientists tested the new antibiotic on mice with the same bladder infection. They divided these mice into three groups, **C**, **R** and **A**.

- Group **C** was the control (untreated).
- Group **R** was treated with an antibiotic currently used against this bladder infection.
- Group **A** was treated with the new antibiotic.

They removed samples from the bladder of these mice after treatment and estimated the total number of bacteria in the bladder.

Their results are shown in **Figure 2**.

Figure 2



The antibiotics were given to the mice at a dose of 25 mg kg^{-1} per day.

Calculate how much antibiotic would be given to a 30 g mouse each day.

Show your working.

[2 marks]

Answer = _____ mg

0 5 . 4

Calculate the percentage difference in actual numbers of bacteria in group **A** compared with group **R**. The actual number of bacteria can be calculated from the $\log_{10}$ value by using the 10^x function on a calculator.

Show your working.

[2 marks]

Answer = _____ %

Question 5 continues on the next page

0 5 . 5

The scientists suggested that people newly diagnosed with this bladder infection should be treated with both the current antibiotic and the new antibiotic.

Explain why the scientists made this suggestion.

Use information from **Figure 2** and your knowledge of evolution of antibiotic resistance in bacteria in your answer.

[3 marks]

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ANSWER IN THE SPACES PROVIDED**

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